Familiarizing yourself with Kali

In this lab you'll learn how to install Kali, a Linux based operating system (OS). Linux is an widely used OS found in multiple devices such as phones, cars, fridges, and even servers. It's widely used due to be it being free unlike Window and not to mention it's highly customizable, so you modify it to suit your exact need with no restrictions.

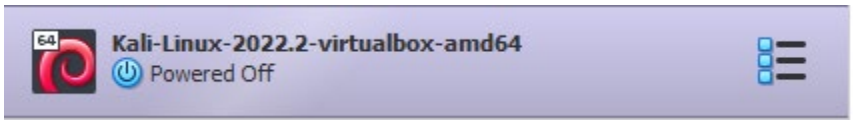
You can use this step-by-step instruction to help setup a Linux Environment on your Windows machine so that you can practice and familiarize yourself with Linux; without having to purchase a second machine exclusively to learn Linux or install Linux onto your main machine if you still prefer Window.

For us in the IT industry, it's very important we familiarize ourselves with Linux because it's widely adopted through the industry. You can consistently find servers built using Linux distribution which might not offer you any GUI. So, it's imperative that you have an environment to practice Linux on. Not to mention a lot of tools used in the Cybersecurity field is optimized to work on Linux.

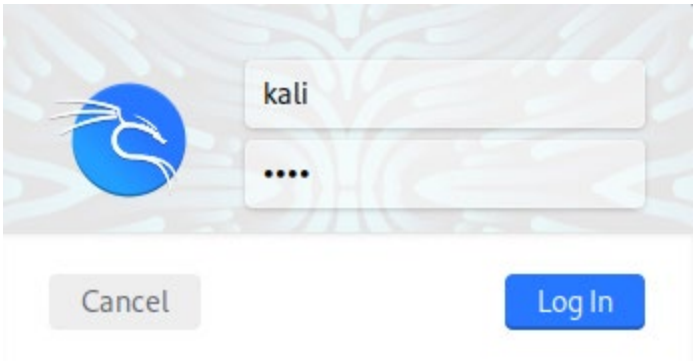
Disclaimer: To follow this lab you must first complete the lab that teach you how to install VirtualBox or previous have VirtualBox install on your machine.

Opening the terminal:

1) Open your Kali virtual machine you made in the previous lab.



2) Press enter on kali and when prompted to put in a username and password, put in kali (this is the default, it's good practice to change it) for both the username and password field.



3) When you log in, on the top left-hand corner, click this icon to open the terminal.

pwd (Print Working Directory)

1) "**pwd**" stands for print working directory and is the command to you used when you want to find which directory or the path to the current directory. As you can see you're currently in home /home/kali, but if you change directory it'll display a different result as seen in later part of this lab.

```
┌──(kali㉿kali)-[~]
└─$ pwd
/home/kali
```

ls (List Files and Directories)

1) "**ls**" (this is a lower case l and not a capital I) lists the files and directories that is in your currently directory with exception of some hidden files.

```
┌──(kali㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

   i) You can do "**ls <directory>**" without the <> to view what's inside the directories. For my purpose. I've created a file on my desktop, so I have a folder called Test File, but you most likely won't

```
┌──(kali㉿kali)-[~]
└─$ ls Desktop
'Test File'
```

mkdir (Make Directory)

1) Now you might be wondering how do I create a file or directory. Well to create a directory you use the command "**mkdir <name>**" such as "mkdir NPower"

```
┌──(kali㉿kali)-[~]
└─$ mkdir NPower

┌──(kali㉿kali)-[~]
└─$ ls
Desktop    Downloads  NPower    Public     Videos
Documents  Music      Pictures  Templates
```

i) As you can see, after I created NPower directory. When I used "**ls**" it now shows the NPower folder.

cd (Change Directory)

1) "**cd**" let you change/move your current directory into one of your choosing. Just use the "**ls**" command to list all the directory currently available to you and then move into one of your choosing.

```
1 ┌──(kali㉿kali)-[~]
  └─$ ls
Desktop    Downloads  NPower    Public     Videos
Documents  Music      Pictures  Templates

2 ┌──(kali㉿kali)-[~]
  └─$ cd NPower

  ┌──(kali㉿kali)-[~/NPower]
  └─$
```

i) As you can see, after moving into NPower. It now lists whatever directory you're in, in the front.

echo

1) "**echo**" is a command that print whatever string you put as an output. So if you put "**echo** Hello World!" it will display this.

```
┌──(kali㉿kali)-[~/NPower]
└─$ echo Hello World!
Hello World!
```

2) You can also use "**echo**" to write to a file by using **>**. For example, "**echo** Hello World! **>** one.txt"

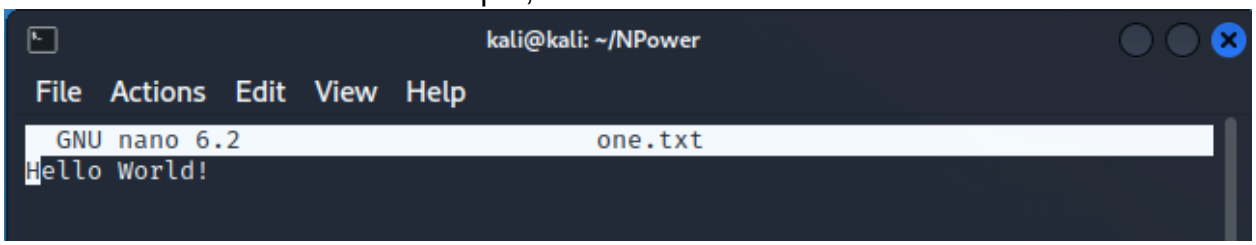
```
┌──(kali㉿kali)-[~/NPower]
└─$ echo Hello World! > one.txt

┌──(kali㉿kali)-[~/NPower]
└─$
```

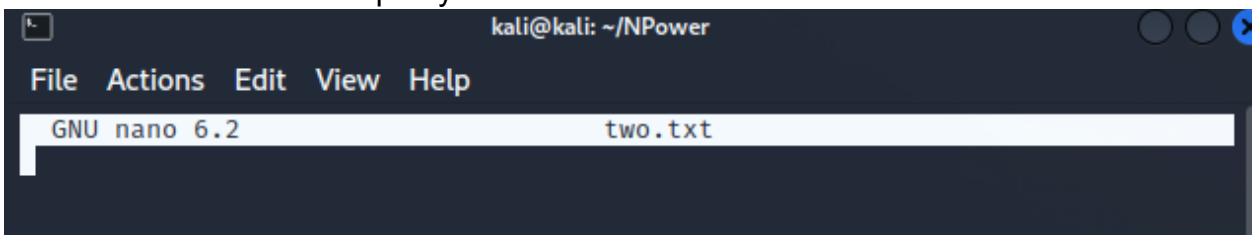
i) As you can see, using **>** there echo didn't display Hello World! because it was instead written into the file called one.txt

nano

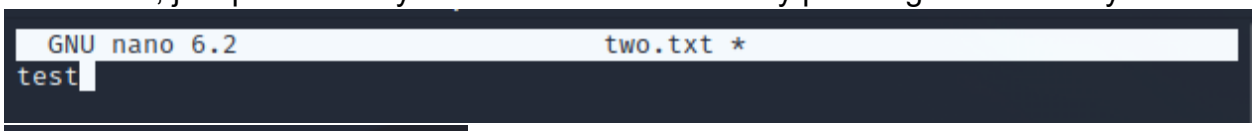
1) “**nano**” is a command that opens the nano editor which allows you to view and edit the content of a file. “**nano** one.txt” opens up the one.txt file we previously created and we can see our input, “Hello World!” inside it.

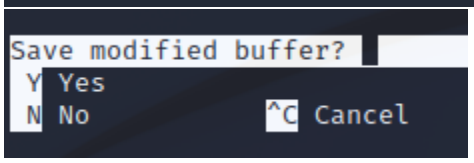


   i) To leave hold down control key and press x.
2) “**nano”** also let you create new files if you so wise. “**nano** two.txt” creates a text file called two.txt and open you into the nano editor.



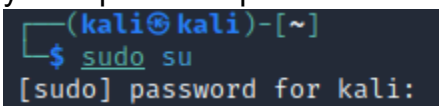
3) If you make any change in two.txt such as typing the word “test” It’ll ask you want to save, just press Y for yes or N for no. Followed by pressing the Enter key





sudo su

1) “**sudo su**” is a command that allows you to make yourself the root user which is the Linux equivalent of administrator on Window. So inputting sudo su will require you to put in the password for kali will be kali



   i) When inputting your password, nothing will show up. This is intentional and is a security measure.
2) After putting in your password you’ll see that you’ve become a user root user when it says root@kali

```
┌──(root㉿kali)-[/home/kali/NPower]
└─#
```

3) To leave root and you simply type “**exit**”

```
└─# exit

┌──(kali㉿kali)-[~/NPower]
└─$
```

ifconfig

1) “**ifconfig**” is the Linux equivalent of ipconfig and it’s used to find your ip information

```
┌──(kali㉿kali)-[~/NPower]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fedb:966a  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:db:96:6a  txqueuelen 1000  (Ethernet)
        RX packets 1  bytes 590 (590.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 26  bytes 3256 (3.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

rm

1) “**rm**” is used to removed files and directories. To just remove files, you simply type “**rm** <file>” without the brackets. For example, “**rm** two.txt” removes two.txt

```
┌──(kali㉿kali)-[~/NPower]
└─$ rm two.txt

┌──(kali㉿kali)-[~/NPower]
└─$ ls
one.txt
```

   i) As you can see, I’ve removed two.txt so when you do ls, you can only see one.txt

2) To remove a directory, we'll first have to leave NPower directory since this one doesn't have any directory. To do so we use "**cd** .." the two .. signify to return to previous directory

```
┌──(kali㉿kali)-[~/NPower]
└─$ cd ..

┌──(kali㉿kali)-[~]
└─$ ls
Desktop    Downloads  NPower    Public     Videos
Documents  Music      Pictures  Templates
```

3) Now we're back at the home directory, we can see other directories. If you try to do "**rm** NPower" you'll see this text.

```
┌──(kali㉿kali)-[~]
└─$ rm NPower
rm: cannot remove 'NPower': Is a directory
```

4) Instead, you have to do "**rm -r** NPower" to remove directories.

```
┌──(kali㉿kali)-[~]
└─$ rm -r NPower

┌──(kali㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

5) As you see NPower directory has been removed

history

1) If you want to see the list of commands used, you can use the command "**history**" to display a list of all the commands used previously

```
┌──(kali㉿kali)-[~]
└─$ history
    1
    2  pwd
    3  PWD
    4  clear
    5  ls
    6  cd NPower
    7  ls
    8  echo Hello World!
    9  echo Hello World! > one.txt
   10  ls
   11  nano one.txt
   12  nano two.txt
   13  sudo su
   14  ifconfig
   15  rm two.txt
   16  ls
   17  cd ..
   18  ls
   19  rm NPower
   20  rm -r NPower
   21  ls
```